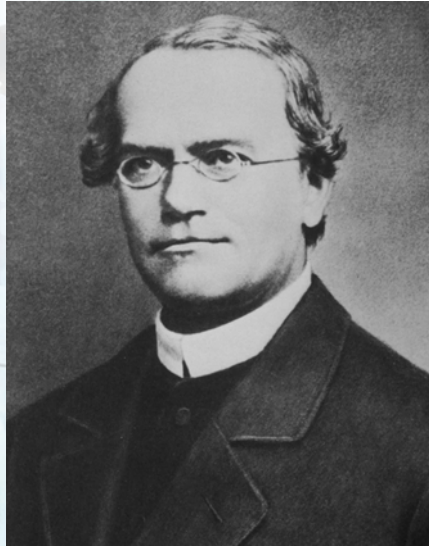




Gregor Mendel.

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Mendel, Gregor

Natural Scientist 1822–1884

Gregor Mendel laid the foundation for the modern understanding of inheritance with his experiments on transmission of traits in garden peas. The ideas he developed are still in use today, and his essential insights into the physical nature of inheritance led directly to the understanding of the gene as a physical entity within the cell.

Education and Training

Mendel was born into a farming family in Heinzendorf, Austria (now Hynčice, Czech Republic). He attended university in Olmutz, but financial difficulties soon persuaded him to enter the Augustinian monastery in Brno, where he received both theological and agricultural training. Mendel remained affiliated with the monastery for the rest of his life. He served briefly as a parish chaplain in the region, and for many years served as a popular and successful teacher at the technical school in Brno. His training in agricultural experimentation, obtained at the University of Vienna, beginning in 1851, prepared him for the experiments that he began in 1856 on peas.

Experiments on Peas

Mendel's experiments were designed to investigate the most widely accepted model of inheritance, blending, which held that the traits of an offspring would be a blend of the parental traits. For example, the theory of blending predicts that a tall and short parent would give rise to a medium-height offspring. Mendel's results showed that for many simple traits, at least, this model was wrong. Instead, the offspring displayed traits in exactly the same form as they appeared in one or the other of the parents.

Mendel chose to study a small group of traits that occur in either of two forms, such as round versus wrinkled pea shape. He began by developing "pure-breeding" lines of each form. In a pure-breeding line, crossing two members gives only offspring that are identical to the parents for that trait. Mendel then crossed pure-breeding parents who had different forms of a trait. For example, he crossed a pea plant that produced only round peas with one that produced only wrinkled peas. All the offspring from this cross developed only round peas; no wrinkled peas were found. When these offspring were crossed among themselves, however, both round and wrinkled were observed, in a numerical ratio of three round-pea plants for every one wrinkled-pea plant.

Mendel explained these results by proposing that each visible trait is governed by the presence of two “factors,” which may be the same or different in any individual. One of these factors is “dominant,” while the other is “recessive.” In the above example, the round-producing factor is dominant, and the wrinkled-producing factor is recessive. If two recessive factors are present, the organism will display the recessive trait. If the organism has two dominant factors, or one dominant and one recessive, the dominant trait will be displayed.

Laws of Inheritance

To explain the numerical relationships he obtained, Mendel developed the Law of Segregation. He proposed that during the process of egg and sperm formation, the two factors separate, or segregate, so that each egg or sperm contain only one factor. For a parent containing one of each type of factor, this means that half the sperm (or eggs) will contain the dominant factor, and half the recessive factor. During fertilization, these randomly pair up, so that some offspring will have two dominants, some two recessives, and some one of each. Simple algebra shows that the ratio of offspring in such a cross will be 3:1, just as Mendel found.

To show how this works, let $0.5D$ be the proportion of dominant factors and $0.5r$ be the proportion of recessive factors. Multiplying $(0.5D + 0.5r)$ times itself gives the offspring ratios, $0.25D^2 + 0.5Dr + 0.25r^2$. In this expression, $0.25D^2$ indicates that one-quarter of the offspring will have both dominant factors, $0.5Dr$ means half will have one of each type, and $0.25r^2$ means one-quarter will have both recessive factors. Since both the D^2 and Dr organisms will show the dominant trait, the ratio of dominant to recessive traits in the offspring will be $0.75:0.25$, or 3:1.

Mendel went on to study crosses between peas with multiple sets of traits, such as round seeds plus tall plants crossed with wrinkled seeds plus short plants. He found that the factors for each trait acted independently, so that the offspring of these crosses showed all possible combinations of traits. From the results of these experiments, he formulated his second principle, known as the Law of Independent Assortment, which states that the members of factor pairs assort (segregate) independently of each other during sperm and egg formation, and combine again randomly.

Mendel's Scientific Legacy

While neither Mendel nor anyone else in his day knew anything about chromosomes or genes, the laws of inheritance he discovered predicted exactly how genes behave on chromosomes during the reproductive process. Indeed, the factors he discovered are genes, which come in pairs and segregate on separate chromosomes during sperm and egg production, just as he suggested. Gene pairs located on different sets of chromosomes will assort independently during the process. While most genes do not exhibit simple dominance-recessiveness relations, and most traits are governed by more than one gene, it is to Mendel's credit that he began by trying to understand simple systems in order to develop generalizable laws.

Mendel published the results of his experiments, “Versuche über Pflanzenhybriden” (“Attempts at Plant Hybridization”) in 1866. He did little scientific work after he became abbot of the monastery two years later.



His work was ignored by the larger scientific community, in part because it was not published in a widely read journal, and in part because it tackled a problem, the physical basis of heredity, that few other scientists were thinking deeply about at that time.

That changed shortly afterward, when microscopic studies of cells revealed that chromosomes divided when cells divided, provoking speculation that they might be involved in inheritance. Mendel's studies were rediscovered in 1900, sixteen years after his death, by three biologists studying similar phenomena. The importance of his theory of inheritance was immediately recognized and widely accepted, and became the starting point for further investigations of the nature of inheritance that were carried out by Thomas Hunt Morgan, Alfred Sturtevant, and other twentieth-century geneticists. Mendelism, as the theory was called, was merged with Darwinism in the 1930s to form the "New Synthesis," which explained evolutionary theory in modern genetic terms. SEE ALSO CHROMOSOMAL THEORY OF INHERITANCE, HISTORY; INHERITANCE PATTERNS; MENDELIAN GENETICS; NATURE OF THE GENE, HISTORY; PROBABILITY.

Richard Robinson

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Mendelian Genetics

hybrids combinations of two different types

Gregor Mendel (1822–1884), an Austrian monk and botanist, was curious and loved nature. He grew plants with diverse flower colors, and he cross-pollinated plant varieties to create **hybrids**. Mendel's fascination with "the striking regularity with which the same hybrid forms always reappeared," broadened his quest into discovering laws for inheriting any trait, not just flower color, from one generation to the next.

Mendel designed a series of experiments to learn the statistical rules governing the features that appeared in hybrids and in their offspring. Mendel identified plant varieties that exhibited the same features over many generations when the plants were allowed to self-pollinate or cross-pollinate with plants from the same variety. He chose hybrids that were fertile, so that their inherited characteristics, or traits, could be passed on to their offspring. He also made sure to exclude foreign pollen, so that outside plants did not get mixed up in his breeding experiments. Mendel chose peas as an ideal plant that had these characteristics.

Mendel obtained thirty-four varieties of peas from seedsmen, and, after two years of preparative work, he selected for study seven traits exhibited by the peas. The seven traits were: color of the seed coats (white or non-white); form of the ripe seeds (round or wrinkled); color of the seeds (yellowish orange or green); form of the ripe pods (inflated or constricted); color of the unripe pods (dark green or vivid yellow); position of the flowers (axial or terminal); and length of the stems (long or short).

Mendel carefully avoided choosing any traits, such as size and form of leaves, length of flower stalk, or size of pods, that would have generated a

		maternal gametes			
		AB	Ab	aB	ab
paternal gametes	AB	AABB	AABb	AaBB	AaBb
	Ab	AABb	AAbb	AaBb	Aabb
	aB	AaBB	AaBb	aaBB	aaBb
	ab	AaBb	Aabb	aaBb	aabb

Figure 1. This Punnett square shows the possible outcomes, in genotypes, of a dihybrid cross (AaBb x AaBb). Reginald Punnett designed this layout to illustrate how alleles will assort according to Mendel's laws (segregation and independent assortment). The two parents are heterozygous for two traits, A and B, on non-homologous chromosomes and carry the genotype AaBb. The four gamete types from the male and female are shown in boldface. Lowercase letters represent recessive alleles; uppercase letters represent dominant alleles.

chaotic mix of forms. He chose traits that would allow plants and their offspring to be distinctly classified.

Instead of looking at all seven traits at once, Mendel focused on one at a time. For each trait, he crossed two plant varieties to make hybrid plants. This was a monohybrid cross, because only one of the plant's many traits was studied. Mendel crossed the two chosen forms for each of the seven traits, using several hundred plants in each cross.

He found that in each case, all the first-generation offspring exhibited the same form as one of the parents, despite the hybrid having received input from two different parental varieties. Mendel called the form of the trait that appeared in these first-generation offspring **dominant**, as it was able to hide the other form during that generation. When the first-generation hybrid plants were allowed to self-pollinate, the hidden feature resurfaced in the next generation. Mendel called the hidden feature **recessive**. He further discovered that, on average, for every four offspring in the second generation, three displayed the dominant form of the trait, and one displayed the recessive form. He used these observations to suggest that each trait was governed by two "factors," one dominant and one recessive.

Mendel concluded that each plant carried two factors for every trait—either two dominant factors, two recessive factors, or one dominant and one recessive factor.

He proposed that his true-breeding parents carried two factors of the same kind. This is now defined as being homozygous. One parent plant was homozygous dominant, and the other homozygous recessive. When the parents were crossed, each offspring plant inherited one factor from each parent, but exhibited only the dominant form of the trait, even though they had received both a dominant and a recessive form. The offspring plants were hybrids, now called **heterozygotes**.

dominant controlling the phenotype when one allele is present

recessive requiring the presence of two alleles to control the phenotype

heterozygotes individuals whose genetic information contains two different forms (alleles) of a particular gene

When these heterozygous plants self-pollinated, their offspring had an equal chance of receiving either two recessive factors, two dominant factors, or a dominant and a recessive factor. One quarter of these offspring were homozygous recessive, one quarter were homozygous dominant, one-half, were heterozygotes. Except for the one quarter that were homozygous recessives, the rest had at least one dominant factor and showed the dominant form of the trait. This explained Mendel's observation that three of every four plants showed the dominant form, and one in four the recessive.

Mendel also allowed the offspring of the heterozygous plants to self-pollinate. When he let plants with recessive features self-pollinate, only recessive features developed in their descendants, supporting the theory that they all contained only recessive factors.

When he let plants with dominant features self-pollinate, one-third gave rise to descendants that exhibited only dominant features. The other two-thirds gave rise to progeny with both dominant and recessive features, and therefore had to contain both dominant and recessive factors. Mendel tested six generations of plants and got similar results for each generation. Each generation of self-pollinating heterozygotes bore offspring, of which half were heterozygotes and half were homozygotes.

Mendel also did reciprocal crosses for each of the seven traits, switching the egg-bearing and the pollen-bearing variety to transmit the dominant and recessive features. The same ratio—three plants with dominant features for every one with recessive features—emerged from all the reciprocal crosses. Mendel concluded that a descendant had an equal chance of getting a dominant or a recessive factor (now called alleles) from either **heterozygous** parent, regardless of sex.

The Principle of Segregation

Mendel used his observations to formulate his First Law, the Principle of Segregation. According to this principle, each **gamete** receives from a parent cell only one of the two alleles the parent cell carries for each trait, and the gamete has an equal chance of getting either allele. (Exceptions to the principle were described later by Thomas Hunt Morgan, an American geneticist.) When the two gametes unite during fertilization, the resulting cell contains two alleles, either identical or different, for each trait. These two alleles are referred to as the individual's genotype for the trait.

An alternative idea that other scientists during Mendel's time had was that two parental characteristics fused and blended into a single hybrid characteristic. Mendel's results showed this was not the case. His results showed, instead, that individuals inherit from their parents intact units that can leap through time. Mendel's discovery that inheritance had a particulate nature set the stage for modern advances in genetics.

The Principle of Independent Assortment

Mendel began a numerical evaluation of, respectively, two and three traits simultaneously, because he also wanted to know how different traits sorted themselves during gamete formation. He began a numerical evaluation of how two and then three traits were inherited simultaneously. The dihybrid cross involved two traits: the form of the plant's ripe seeds and the color of

heterozygous characterized by possession of two different forms (alleles) of a particular gene

gamete reproductive cell, such as sperm or egg

its interior seeds. He crossed one true-breeding variety that had wrinkled seeds and a green interior, with another that had round seeds and a yellow interior. The dihybrid cross generated offspring that all had round, yellow seeds, but the seeds' outward appearance, or phenotype, hid the offspring's heterozygous nature. The offspring contained recessive alleles for making wrinkled, green seeds, as well as the dominant alleles that generated the seeds' round and yellow appearance.

The round, yellow seeds, which were the seeds of the first filial generation, or F1, were planted, raised, and made to self-pollinate. Their progeny, the second filial generation, or F2, had four phenotypes for seed form and color, in a ratio of 9:3:3:1 (nine round and yellow, to three wrinkled and yellow, to three round and green, to one wrinkled and green).

To unmask the F2 genotypes, the next generation's wrinkled, round, yellow, or green seeds were collected. The seeds showed that there were nine different genotypes among the F2 plants. If Y represents yellow, y represents green, R represents round, and r represents wrinkled, the nine genotypes were: YyRr, YyRR, Yyrr, YYRr, YYRR, YYrr, yyRr, yyRR, and yyrr. Four of the genotypes were homozygous for both traits, four were homozygous for one trait and heterozygous for the other, and one was heterozygous for both traits.

Mendel's trihybrid cross included the trait for the color of the seed coats, which could be white or non-white, in addition to the same two traits used in the dihybrid cross. In this cross, the F2 generation had eight different combinations of seed shape, seed coat color, and interior seed color and twenty-seven different genotypes.

The existence of all these allelic combinations revealed that chance had a lot to do with what ended up in the same gamete. The chance of a descendant getting a specific seed shape and color depended on straight math, not on interaction between shape and color or another unknown influence. A ratio of three dominant to one recessive phenotype appeared for each trait, as if the other traits' alleles did not exist. The arrival of one allele inside a gamete was unaffected by the entry of another trait's allele. Mendel described this formally as "each pair of different characters in hybrid union is independent of the other differences."

The chance of a descendant getting a specific trait depends on probability, not on the interaction between traits. This is formally stated as Mendel's Second Law, or the Principle of Independent Assortment: Different traits assort (i.e. are included in gametes) independently of one another.

A Punnett square, designed by English geneticist Reginald Punnett (1875–1967) and shown in Figure 1, shows the outcomes of crosses that follow Mendel's laws. The capital letters A and B represent dominant alleles, and the lowercase letters a and b represents recessive alleles. A genotype that is heterozygous for both traits in a dihybrid cross is represented as AaBb.

Exceptions to Mendel's Laws

The seven traits that Mendel evaluated all assort independently, but not all sets of traits do. Independent assortment is true for the seven traits that Mendel evaluated, and holds generally true for traits (genes) found



on non-homologous chromosomes. Any chromosome carries a collection of traits located on a long string of DNA, and the traits are therefore physically linked in a series or sequence. A non-homologous chromosome carries a unique collection of traits on a long string of DNA, that is different from the gene collection of an other non-homologue. Normal non-homologous chromosomes are not attached to each other during meiosis, and move independently of one another, each carrying their own gene collection. Each chromosome, composed of a long string of DNA, carries a collection of genes, with each gene showing up in a particular form or type. Seven chromosomes reside within a pea gamete, and each of the traits Mendel chose to study lie on a different (non-homologous) chromosome.

Independent assortment is not true for the collection of traits that are located on a homologous chromosome. In eukaryotes, homologues come in pairs, one donated from each parent. Two homologous chromosomes carry the same collection of genes, but each gene can be represented by a different allele on the two homologues (a heterozygous individual). A gamete will receive one of those homologues, but not both. Genes or alleles that travel together on a chromosome do not show independent assortment, because they do not move independently of each other into a gamete.

Punnett and William Bateson (1861–1926), an English biologist, published the first report of gene linkage in peas. A comparison between the ratios at which certain genes were inherited and the expected Mendelian ratios showed that the traits did not assort independently.

Sometimes two traits on non-homologous chromosomes affect each other's phenotypic expression. Purple flowers, for example, occur only with the presence of at least one dominant allele from two different genes. Offspring of two parents who are heterozygotes for both genes produce flowers that are purple or white at a ratio of nine to seven. The genotypes of the purple offspring are either $aaB-$ or $A-bb$. (A dash indicates that the individual could have either a dominant or a recessive allele.)

Incomplete dominance occurs when a heterozygote has a unique phenotype. Pink flowers, for example, result when one parent is homozygous white and the other homozygous red. Neither allele hides the other, and their appearance together creates a unique intermediate phenotype.

Alleles are said to be codominant when heterozygotes express both alleles but neither affects the other's character. Individuals who have the allele for blood type A and the allele for blood type B, for example, have the characteristics of both blood types and are referred to as being of blood type AB.

Modern studies of genetic diseases use Mendel's ratios to determine whether or not genes are linked to certain chromosomes. Family histories are converted into pedigrees to help understand inheritance patterns. A disease might skip generations as expected of recessive alleles, or be linked to other traits. Deafness, white hair, and blue eyes are linked in cats, for example. A disease's symptoms might also become more severe with successive generations, as is the case with some dominant alleles.

Mendelian genetics and molecular biology together can elucidate the function of genes that are critical for development and life, in both experimental animals and human beings. Understanding of genetic processes can

help to cure diseases. SEE ALSO CHROMOSOMAL THEORY OF INHERITANCE, HISTORY; INHERITANCE PATTERNS; MEIOSIS; MENDEL, GREGOR; MORGAN, THOMAS HUNT; NATURE OF THE GENE, HISTORY; PROBABILITY.

Susanne D. Dyby

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Metabolic Disease

Metabolism is the sum of the chemical processes and interconversions that take place in the cells and the fluids of the body. This includes the absorption of nutrients and minerals, the breakdown and buildup of large molecules, the interconversion of small molecules, and the production of energy from these chemical reactions. Virtually every chemical step of metabolism is catalyzed by an **enzyme**. Disorders of these enzymes that result from abnormalities in their genes are known as inborn errors of metabolism.

Inborn errors of metabolism were first recognized by Sir Archibald Garrod, a British physician who noted in 1902 that the principles of Mendelian inheritance applied to certain examples of human metabolic variation. He perceived the genetic basis for a particular metabolic condition that leads to visible effects—alcaptonuria, which results in a black pigment in the urine. Since then, more advanced chemical methods have allowed the discovery of hundreds of enzyme defects that cause metabolic diseases.

Enzymes Control Metabolic Reactions

Enzymes are proteins that control the rate of chemical reactions in the cell. In general, each enzyme controls the rate of only one or a few reactions. Enzymes function by binding to the molecules to be reacted (called substrates or precursors) and altering their chemical bonds, producing products. The binding occurs on the surface of the enzyme, usually in a pocket or groove, called the active site. The enzyme releases the products after reaction. The active site has a specific three-dimensional structure that is required for binding substrates. In addition, it may have other sites that bind regulatory molecules or cofactors. Some cofactors are vitamins, which perform some accessory function critical for enzyme action.

Enzymes are often linked in multistep pathways, such that the product of one reaction becomes the substrate for another. In this way, a simple molecule can be changed step by step into a complex one, or vice versa. In addition, the multiple steps provide additional levels of regulation, and intermediates can be shunted into other pathways to make other products. For instance, some intermediates in the breakdown of sugar can be shunted to make amino acids. When all the enzymes in a pathway are functioning properly, intermediates rarely build up to high concentrations.

enzyme a protein that controls a reaction in a cell



William Lucas, shown in 1998, when he was twelve, must wear a protective suit while riding his bicycle because he suffers from a metabolic disease called Erythropoietic Protoporphyrria (EPP). The disorder is characterized by extreme sunlight sensitivity that can cause exposed skin to swell and blister.



Enzyme Defects Cause Metabolic Disorders

The causes of enzyme defects are genetic mutations that affect the structure or regulation of the enzyme protein or create problems with the transport, processing, or binding of cofactors. In general, the consequences of an enzyme deficiency are due to perturbations of cellular chemistry, because of either a reduction in the amount of an essential product, the buildup of a toxic intermediate, or the production of a toxic side-product, as shown in Figure 1.

Except as noted below, most metabolic disorders are inherited as autosomal recessive conditions. In this inheritance pattern, two defective gene copies are needed (one from each parent) to develop the disease. The parents, each of whom almost always has only one gene copy, will not have the disease but are **carriers**. The chance that two carrier parents will have a child who inherits two defective gene copies is 25 percent for each birth.

Metabolic disorders tend to be recessive, because they are due to inactivating, or “loss-of-function,” mutations. One working copy of the gene is

carriers people with one copy of a gene for a recessive trait, who therefore do not manifest the trait

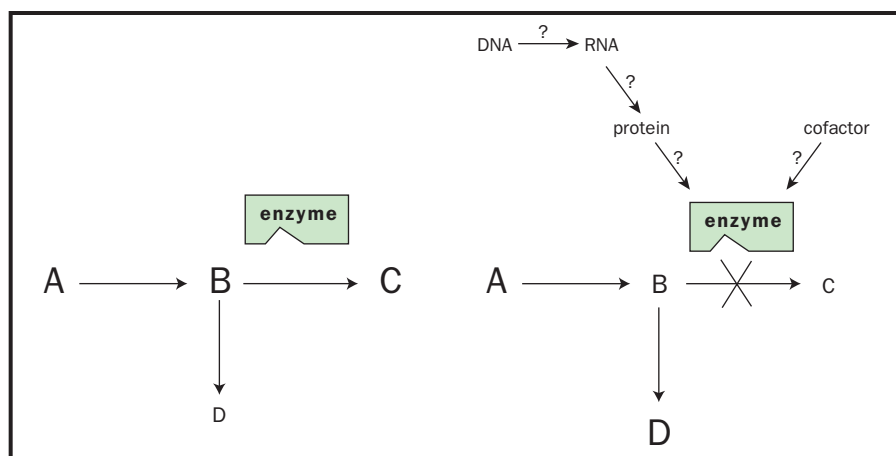


Figure 1. *Left:* Normal sequence of metabolism, in which precursor A is converted to intermediate B, and then into final product C. The B-to-C conversion is catalyzed by the enzyme. The side-product D is made in very small amounts. *Right:* Metabolic disease, resulting in altered enzyme*, resulting in decrease of product C, and/or buildup of precursor A and/or side-product D. The enzyme defect may be due to either an abnormality of the gene (DNA), a problem with production of messenger RNA (mRNA), a defect in the production or stability of the protein, or a deficiency or abnormal interaction with an enzyme cofactor.

usually enough to maintain sufficient levels of the enzyme, and so with one copy present, no disease develops.

Approaches to Treatment

Treatment approaches for metabolic disorders include (a) modifying the diet to limit the amount of a precursor that is not metabolized properly; (b) using cofactors or vitamins to enhance the residual activity of a defective enzyme system; (c) using detoxifying agents to provide alternative pathways for the removal of toxic intermediates; (d) enzyme replacement, to provide functional enzymes exogenously (from the outside); (e) organ transplantation, which in principle allows for endogenous (internal) production of functional enzymes; and (f) gene therapy, or replacement of the defective gene.

Gene therapy is expected to become the most important approach. It offers the potential for definitive treatment, and it is being actively investigated as a treatment for virtually every one of the metabolic disorders. Most of the genes for the enzymes involved in metabolic diseases have been identified and cloned, and in many cases the genes can be replaced in experimental systems. Genetic approaches have been used to produce mass quantities of enzymes to use for enzyme replacement, but as of 2002, gene therapy has not yet been used successfully to provide the stable expression of active enzymes in the human body.

This chapter will summarize classes of inborn errors of metabolism based upon the type of chemical process involved, and individual disorders will be discussed that illustrate the various disease mechanisms and treatment approaches.

Major Classes of Metabolic Disorders

Cells are constructed from four major types of molecules: carbohydrates, proteins, fats, and nucleic acids. The metabolic pathways involving each are

Disease	Defective Enzyme or System	Symptoms	Treatment
Disorders of Amino Acid Metabolism			
Phenylketonuria (PKU)	phenylalanine hydroxylase	severe mental retardation	screening; dietary modification
Malignant PKU	biopterin cofactor	neurological disorder	—
Type 1 tyrosinemia	fumarylacetoacetate hydrolase	nerve damage, pain, liver failure	liver transplantation; preceding enzyme inhibitor plus dietary modification
Type 2 tyrosinemia	tyrosine aminotransferase	irritation to the corneas of the eyes	diet with reduced phenylalanine and tyrosine content
Alkaptonuria	disorder of tyrosine breakdown	progressive arthritis and bone disease; dark urine	—
Homocystinuria and Hyperhomocysteinemia	cystathionine- β -synthase or methylenetetrahydrofolate reductase or various deficiencies in formation of the methylcobalamin form of vitamin B ₁₂	hypercoagulability of the blood; vascular eposides; dislocation of the lens of the eye, elongation and thinning of the bones, and often mental retardation or psychiatric abnormalities	vitamin B ₁₂ , folic acid, betaine, a diet limited in cysteine and methionine
Maple Syrup Urine disease	branched-chain ketoacid dehydrogenase complex	elevations of branched-chain amino acids, characteristic odor of the urine, episodes of ketoacidosis, death	thiamine; careful regulation of dietary intake of the essential branched-chain amino acids
Disorders of Organic Acid Metabolism			
Propionic Acidemia	propionyl-CoA carboxylase	generalized metabolic dysfunction; ketoacidosis; death	diet with limited amounts of the amino acids which are precursors to propionyl-CoA
Multiple Carboxylase deficiency	pyruvate carboxylase and 3-methylcrotonyl-CoA carboxylase	—	biotin
Methylmalonic Acidemia	methylmalonyl-CoA mutase; defects in the enzyme systems involved in vitamin B ₁₂ metabolism	—	supplementation with large doses of vitamin B ₁₂ ; diet
Disorders of Fatty Acid Metabolism			
Hyperlipidemia and hypercholesterolemia	regulation or utilization of lipoproteins	cardiovascular disease	dietary modifications and use of drugs that inhibit fatty acid synthesis.
Fatty Acid Oxidation disorders	very long chain acyl-CoA dehydrogenase; long chain hydroxyacyl-CoA dehydrogenase; medium chain acyl-CoA dehydrogenase; short chain acyl-CoA dehydrogenase; short chain hydroxyacyl-CoA dehydrogenase	low blood sugar (hypoglycemia); muscle weakness; cardiomyopathy	avoidance of fasting, intravenous glucose solutions; carnitine; medium chain triglycerides
Glycogen Storage diseases	defects in glycogenolysis	liver enlargement or damage; muscle weakening or breakdown; disturbed renal tubular function; risk of brain damage	—
Galactosemia	galactose-1-phosphate uridyl transferase	liver failure in infancy	newborn screening; milk avoidance
Congenital Disorders of Glycosylation	defects in the enzymes that build the carbohydrate side-chains on proteins	quite variable; multisystem	—
Disorders of Purine and Pyrimidine Metabolism			
Purine Overproduction	imbalance between purine synthesis and disposal	gout	—
Lesch-Nyhan syndrome	hypoxanthine phosphoribosyl-transferase	defective salvage of purines; increase in the excretion of uric acid; brain neurotransmitter dysfunction; severe spastic movement disorder; self-injurious behavior	allopurinol (does not treat neurological symptoms)
Lysosomal Storage Disorders			
Gaucher disease Types I and II	cerebrosidase	enlargement of the spleen and liver; painful and crippling effects on the bones; severe brain disease and death (Type II)	enzyme replacement (Type I)
Tay-Sachs disease	beta-hexosaminidase A	neurological disorders; enlarged head; death in early childhood	—

Table 1 (continued on next page).

Disease	Defective Enzyme or System	Symptoms	Treatment
Lysosomal Storage Disorders [CONTINUED]			
Fabry disease	α -galactosidase	severe pain; renal failure; heart failure	enzyme replacement
Hurler syndrome, Hunter syndrome	α -iduronidase (Hurler syndrome); iduronate sulfatase (Hunter syndrome)	enlargement of the liver and spleen; skeletal deformities; coarse facial features; stiff joints; mental retardation; death within 5–15 years	enzyme replacement
Sanfilippo syndrome	enzymes for heparan sulfate degradation	enlargement of the liver and spleen	enzyme replacement
Maroteaux-Lamy syndrome	arylsulfatase B	progressive, crippling and life-threatening physical changes similar to Hurler syndrome, but generally with normal intellect	—
Morquio syndrome	galactose 6-sulfatase; β -galactosidase	truncal dwarfism; severe skeletal deformities; potentially life-threatening susceptibility to cervical spine dislocation; valvular heart disease	—
Disorders of Urea Formation			
	carbamyl phosphate synthetase deficiency; ornithine transcarbamylase deficiency, citrullinemia, argininosuccinic aciduria	hyperammonemia; mental retardation; seizures; coma; death	limitation of dietary protein; phenylacetate; liver transplantation
Disorders of Peroxisomal Metabolism			
Refsum disease	branched-chain fatty acid buildup	neurologic symptoms	—
Alanine-glyoxylate transaminase defect	alanine-glyoxylate transaminase	oxalic acid increase; organ dysfunction; renal failure	liver transplantation

Table 1, continued.

the basis for classification of many of the metabolic disorders. The mitochondria in cells are organelles that play a major role in most metabolic pathways, and mitochondrial disorders are one of the most significant and common types of metabolic disorders. Defects in the storage and disposal of molecules also give rise to metabolic disorders.

Carbohydrates are used primarily as fuel and can be built and broken down rapidly. The major storage form is glycogen. They are also added to proteins to make glycoproteins. Fatty acids are long-chain molecules that are used to construct membranes. Fatty acids are derived from dietary fats. Excess fat is used as fuel by mitochondria. Proteins are made of amino acids.

Humans must eat eight kinds of amino acids and then convert these into twelve other types to make the twenty amino acids found in our proteins. Excess amino acids in the diet are used for fuel by mitochondria. Along the way, they generate organic acids. Nucleic acids—DNA and RNA—are the molecules that store and process genetic information. They must be built from smaller units, called nucleotides. The storage and interconversion of different types of nucleotides assures a steady supply.

Below, representative disorders of each system are discussed. Other disorders are listed in Table 1. Many of the disease names end in “emia.” This suffix indicates a blood disorder, and the names are derived from the fact that most metabolic disorders are diagnosed by detecting abnormal levels of intermediates or other substances in the blood.

Disorders of Mitochondrial Oxidative Metabolism

Most cellular energy is derived from the mitochondrial electron transport chain, which reduces oxygen to water in a series of steps to drive the

neurological related to
brain function or dis-
ease

formation of the high-energy compound ATP. The Krebs cycle creates high-energy intermediates that it feeds to the electron transport chain, the energy of which ultimately is derived from a two-carbon compound called acetate, which is broken down successively to carbon dioxide. Acetate is derived from several pathways of amino acid, carbohydrate, and fat metabolism.

Thus, many pathways of metabolism feed into the Krebs cycle to drive oxidative metabolism in a web of processes requiring hundreds of enzymes. When there are defects in the Krebs cycle or the electron transport chain, one result may be ketoacidosis, which is due to the accumulation of lactic acid and ketone bodies.

The lack of cellular energy may be manifest in many cellular processes and can affect several tissues and organ systems, particularly those that are most dependent upon oxidative metabolism for energy. The brain and muscles are generally affected first, which can cause developmental delay, **neurological** crises—including episodes of coma, stroke-like events, and seizures—and muscle weakness or cardiomyopathy. Kidney function—most often the tubular function required for retention of electrolytes—may also be affected. Endocrine (hormone) systems may also be affected, resulting in conditions such as diabetes mellitus (caused by effects on the pancreas or by sensitivity to insulin in muscle and fat cells) or adrenal insufficiency (from effects on the adrenal glands).

Disorders of mitochondrial oxidative metabolism are very variable in terms of age of onset, severity, specific symptoms, and clinical course. Even the inheritance patterns of mitochondrial diseases are heterogeneous. Most are inherited in the usual autosomal recessive manner (although the chromosomal locations of only a few of the relevant genes are known). A few are inherited from defects in the mitochondrial DNA, which is passed on in the maternal line.

The mitochondrion contains a circular chromosome of about 16,500 bases. It codes for thirteen components of the electron-transport chain, as well as transfer RNA molecules and ribosomal RNAs required for their expression. Since there are multiple copies of mitochondrial DNA and there may be mixtures of normal and abnormal mitochondrial DNA (a phenomenon known as heteroplasmy), the precise proportion of mutated mitochondrial DNA may vary in an unpredictable manner from individual to individual within a family, and from tissue to tissue within an individual. There may also be variations within an individual tissue over time, adding to the unpredictability of mitochondrial disease and the difficulty in the diagnosis.

Disorders of Amino Acid Metabolism

Phenylketonuria. Phenylketonuria (PKU) is the most common disorder of amino acid metabolism, and it is a paradigm for effective newborn screening. Phenylalanine is an essential amino acid (meaning that it cannot be synthesized but must be taken in through the diet). The first step to its breakdown is the phenylalanine hydroxylase reaction, which converts phenylalanine to another amino acid, tyrosine. A genetic defect in the phenylalanine hydroxylase enzyme is the basis for classical PKU. Untreated PKU results in severe mental retardation, but PKU can be detected by

screening newborn blood spots, and the classical form can be very effectively treated by using medical formulas that are limited in their phenylalanine content.

The hydroxylase enzyme requires a cofactor called bipterin, which is also a cofactor for other enzymes. Defects affecting the production of bipterin result in another form, so-called malignant PKU. In this form, the other bipterin-dependent hydroxylases are also affected, resulting in deficient neurotransmitter synthesis and significant neurological symptoms.

Alkaptonuria. Alkaptonuria is a disorder of tyrosine breakdown. The intermediate that accumulates, called homogentisic acid, can **polymerize** to form pigment that binds to cartilage and causes progressive arthritis and bone disease and that also is excreted to darken the urine—the effect that allowed Garrod to recognize the genetic inheritance of this inborn error of metabolism.

polymerize to link together similar parts to form a polymer

Disorders of Organic Acid Metabolism

Propionic Acidemia. Propionyl-CoA is formed mainly from the breakdown of four essential amino acids (isoleucine, valine, threonine, and methionine). Defects of the enzyme propionyl-CoA carboxylase result in propionic acidemia, a life-threatening disease characterized by episodes of generalized metabolic dysfunction and ketoacidosis. The basis of treatment is a carefully applied diet containing limited amounts of the amino acids that are precursors to propionyl-CoA.

Methylmalonic Acidemia. Methylmalonyl-CoA is the product of propionyl-CoA carboxylase. There are a variety of metabolic defects in the further metabolism of this compound, resulting in methylmalonic acidemia. The best-known of these conditions arises from a defect in methylmalonyl-CoA mutase, the vitamin B₁₂-dependent enzyme that converts methylmalonyl-CoA to succinyl-CoA, which enters the Krebs cycle. There are other conditions resulting in methylmalonic acidemia that are due to defects in the enzyme systems involved in vitamin B₁₂ metabolism. In some cases, supplementation with large doses of vitamin B₁₂ is effective, but in most cases of methylmalonic acidemia, a special diet is required, similar to that used to treat propionic acidemia.

Disorders of Fatty Acid Metabolism

Hyperlipidemia and Hypercholesterolemia. Dietary fats are distributed through the body attached to proteins, in lipoprotein complexes. There are a number of disorders involving the regulation or utilization of lipoproteins, which result in hyperlipidemia and/or hypercholesterolemia, including the common conditions in adults that are associated with cardiovascular disease. Standard treatment approaches include modifying the diet and administering drugs that inhibit fatty acid synthesis.

Disorders of Carbohydrate Metabolism

The most active pathways in carbohydrate metabolism are glycogenolysis (the breakdown of glycogen, a polymerized form of carbohydrate, which is stored primarily in the liver and muscles), which produces glucose and distributes it through the bloodstream, and **glycolysis**, which releases energy

glycolysis the breakdown of the six-carbon carbohydrates glucose and fructose

and produces pyruvate. Pyruvate is a three-carbon molecule that can be converted to acetate and enter the Krebs cycle or form several building-block molecules. The reverse processes are referred to as glycogen synthesis and gluconeogenesis, respectively.

Glycogen Storage Diseases. A number of defects may occur in glycogenolysis, giving rise to the disorders known as glycogen storage diseases. Glycogen storage diseases may affect the liver (enlarging it or damaging it due to increased amounts of glycogen) or muscle (weakening muscle or causing breakdown during times of exercise, due to inadequate glucose production). There may be additional problems, including disturbed kidney tubular function (which causes loss of nutrients and minerals), and there is a risk of brain damage in cases that result in critically low blood sugar.

Galactosemia. Another common disorder of carbohydrate metabolism is galactosemia, which is due to the inability to form glucose from galactose, the sugar that is found in milk. The classic form of galactosemia is due to a deficiency of the enzyme galactose-1-phosphate uridyl transferase, and, if untreated, it presents in the infant with fatal liver failure. Galactosemia is important because newborn screening (conducted by most developed countries on blood spots collected in the first days of life) has been very successful, and simple alteration of the diet (replacing milk with formulas that contain glucose or glucose polymers) has permitted a generation of individuals to survive with quite normal lives and, in general, normal intellect.

Disorders of Purine and Pyrimidine Metabolism

Purines and pyrimidines are chemicals that form the nucleic acids (DNA and RNA). An important purine compound is adenosine triphosphate (ATP), which is used to transfer chemical energy for processes such as biosynthesis and transport. There are several rare defects in the synthesis of purines and pyrimidines. The most common symptom of purine overproduction is gout, which arises for several reasons, often not associated with an identifiable enzyme defect but rather due to an imbalance between purine synthesis and disposal. Gout manifests when the ultimate product of purine degradation, uric acid, accumulates and crystallizes in the joints.

A very dramatic disorder of purine metabolism is Lesch-Nyhan syndrome, which is due to a defect in the enzyme hypoxanthine phosphoribosyltransferase (HPRT), resulting in defective salvage of purines and, accordingly, in an increase in the excretion of uric acid. For reasons that are still incompletely understood, a severe defect of HPRT also causes brain-neurotransmitter dysfunction, resulting in a severe spastic form of movement disorder and also a stereotypical compulsion for self-injurious behavior. The concentration of uric acid can be reduced by using the drug allopurinol, but there is no satisfactory treatment for the neurological symptoms associated with Lesch-Nyhan disease.

Lysosomal Storage Disorders

Lysosomes are intracellular compartments in which **macromolecules** are broken down in an acidic environment. Various classes of lysosomal storage disorders arise when there are defects in specific enzymes, and the manifestations of these disorders depend upon the class of macromolecule whose breakdown is affected.

macromolecules large molecules such as proteins, carbohydrates, and nucleic acids

Gaucher's Disease. The most common lysosomal storage disorder is Gaucher's disease, caused by a deficiency of the enzyme cerebrosidase, which is needed to break down cerebroside, a component of the cell membrane in blood cells and **neurons**. Partial defects of cerebrosidase cause Type 1 Gaucher's disease, in which material accumulates in the lysosomes of **macrophage** cells in the spleen, liver, and bone marrow, where most of the cell-turnover takes place. Significant accumulation usually occurs by childhood or early adulthood, resulting in dramatic enlargement of the spleen and liver. Later there may be painful and crippling effects on the bones. Type 1 Gaucher's disease can be effectively treated with enzyme replacement, but the enzyme must be infused intravenously approximately every two weeks for life. More severe defects of cerebrosidase cause Type 2 Gaucher's disease, which is rare, appears in infancy, and presents with the same problems as in Type 1 disease as well as severe brain disease that progresses to death. Very rarely, defects of intermediate severity can give rise to Type 3 Gaucher disease, which is a chronic neuronopathic form.

Tay-Sachs Disease. Tay-Sachs disease is due to a defect in the beta-hexosaminidase A enzyme, which removes a sugar from certain lipids called gangliosides, which build up in the lysosome. The disease causes neurological symptoms, an enlarged head, and death in early childhood.

Mucopolysaccharidosis. Mucopolysaccharidoses are lysosomal storage disorders affecting the breakdown of mucopolysaccharides, which are carbohydrate-protein macromolecules found on several cell types. Hurler syndrome (α -iduronidase deficiency) and Hunter syndrome (iduronate sulfatase deficiency) are two disorders that affect the breakdown of the mucopolysaccharides dermatan sulfate and heparan sulfate, which are components of connective tissues throughout the body. The usual clinical manifestations of these syndromes are enlargement of the liver and spleen, skeletal deformities, coarse facial features, stiff joints, and mental retardation. Most cases are severe and progress to death within five to fifteen years, but there are exceptions. By 2002, there were several experimental approaches with enzyme replacement for mucopolysaccharidoses.

Disorders of Urea Formation

The urea cycle is a series of enzyme reactions that removes waste nitrogen from the body, allowing it to be excreted in the urine as urea. Disorders of the enzymes of the urea cycle disrupt this pathway, increasing blood ammonia (hyperammonemia). Hyperammonemia results in mental retardation, and acute episodes can progress to seizures, coma, and death. These conditions are inherited in an autosomal recessive pattern, except for ornithine transcarbamylase deficiency, which is X-linked, affecting males more severely than females. Treatment for these disorders includes limiting dietary protein (the major source of nitrogen intake) and using agents (such as phenylacetate) that provide an alternate mechanism to remove waste nitrogen (through excretion of phenylacetyl-glutamine in urine). Liver transplantation may also be effective in controlling blood ammonia in these conditions.

Disorders of Peroxisomal Metabolism

Several specialized metabolic functions are performed in the subcellular organelles known as peroxisomes. Severe defects in the biogenesis of

neurons brain cells

macrophage immune system cell that consumes foreign material and cellular debris



peroxisomes result in Zellweger syndrome, which is characterized by structural and developmental abnormalities and which is generally fatal in infancy. Defects in individual peroxisomal enzymes are also encountered, including Refsum disease, which results in the buildup of a branched-chain fatty acid (phytanic acid) and progressive problems in the nervous system. A defect in the enzyme alanine-glyoxylate transaminase causes an increase in the production of oxalic acid, an insoluble chemical that is progressively deposited in the tissues of the body and, over years, causes organ dysfunction, including renal failure. Renal transplantation does not prevent recurrence, but liver transplantation is effective in preventing the progression of the disease in the kidneys and other organs. SEE ALSO CELL, EUKARYOTIC; INHERITANCE PATTERNS; MITOCHONDRIAL DISEASES; POPULATION SCREENING; PROTEINS; TAY-SACHS DISEASE.

Bruce A. Barshop

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Methylation

DNA methylation is a mechanism used to regulate genes and protect DNA from some types of cleavage. It is one of the regulatory processes that are referred to as epigenetic, in which an alteration in gene expression occurs without a change in the nucleotide sequence of DNA. Defects in this process cause several types of disease that afflict humans.

Biochemical Features

In methylation, a methyl group ($-\text{CH}_3$) is added to position five of the cytosine ring in a DNA molecule (see diagram), attaching itself there by means of a chemical bond. For methylation to occur in DNA, certain conditions must be met. The cytosine must be linked to guanine, with the guanine occurring at the 3' ("three prime") end of the DNA molecule, in a formation that, in scientific notation, is expressed as 5'-CG-3' and is referred to as a CpG dinucleotide (with the "p" representing a phosphate group). It occurs in many eukaryotic organisms, including mammals, and was recently found to occur in *Drosophila* (fruit fly), but does not occur in yeast.

The methylation process is performed by enzymes called DNA methyltransferases (DNMTs). Currently, five DNA methyltransferase members have been identified in humans (DNMT1, 2, 3A, 3B, 3L). The precise function of many of these proteins is not yet known. The most well-characterized DNA methyltransferase is DNMT1. This **enzyme** is required for proper embryonic development in mammals, and is involved in copying the methylation pattern from an existing DNA strand to the newly synthesized DNA strand following DNA **replication**. For this reason, DNMT1 is called the maintenance DNA methyltransferase. In contrast, DNMT3A and DNMT3B are believed to be de novo methyltransferases, or proteins that can add a

enzyme a protein that controls a reaction in a cell

replication duplication of DNA

methyl group to a cytosine at a new location in the DNA strand, instead of just copying one that already exists. It is not yet known what determines which cytosines in the DNA will have a methyl group added by a *de novo* methyltransferase.

CpG Islands

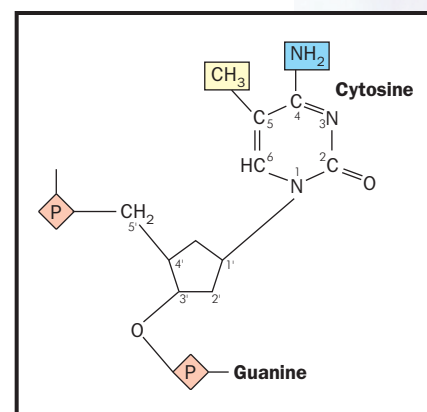
The frequency of occurrence of the CpG dinucleotide in the genome is not random, as would be expected. Instead, the CpG dinucleotide is greatly under represented in eukaryotic genomes, occurring at approximately 5 to 10 percent of its predicted frequency, according to some estimates. Of these occurrences, it is further estimated that 70 to 80 percent are methylated. This under representation of CpG dinucleotides in the genome may result from a spontaneous conversion of methyl cytosine to thymine in DNA by a process known as deamination, in which an amino group (in this case, NH_2) is removed from 5-methylcytosine. For this reason, methylated cytosines represent potential sites of spontaneous DNA mutation in the genome.

There are, however, small regions of DNA that are very rich in linked cytosines and guanines, but which are unmethylated. These regions, which can consist of from 500 to 5,000 base pairs of unmethylated DNA, are referred to as CpG islands. These “islands” commonly occur in **promoter** regions of genes (regions where RNA polymerase binds to start transcription), which are located at the 5′ (“five prime”) end of the genes. In fact, about 50 percent of all genes contain a CpG island in their promoter regions. The lack of methylation in CpG islands leads to a less compact chromatin structure, and generally allows for active gene expression. The methylation of unmethylated CpG islands leads to the silencing of genes required for proper cell growth control and is a common mechanism in the development of many types of cancer.

Host Defense

The process of methylation was first described in bacteria in 1948. Most bacterial strains contain enzymes called restriction **endonucleases**. These restriction enzymes recognize certain short sequences of DNA, and cleave the DNA strand at these sites. By modifying its DNA with a pattern of methylation specific to its strain, a bacterium can use this system of modification and restriction to distinguish its own DNA from invading foreign DNA. Methylation serves to protect the bacterial DNA from digestion by its own restriction enzymes.

In mammals, methylation has also been proposed to be a genome defense system against foreign DNA such as viruses. Viruses that infect cells and integrate into the host cell DNA frequently become methylated. While methylation in eukaryotes does not mark DNA for digestion, methylation can inactivate a promoter and thereby silence gene expression from a viral promoter. Evidence in support of this comes from the fact that most methylated cytosines in the mammalian genome lie within viral and **transposon** DNA. In addition to silencing gene expression from foreign DNA promoters, methylation has also been shown to prevent DNA sequences such as transposons from moving to a new site in the DNA. In this way,



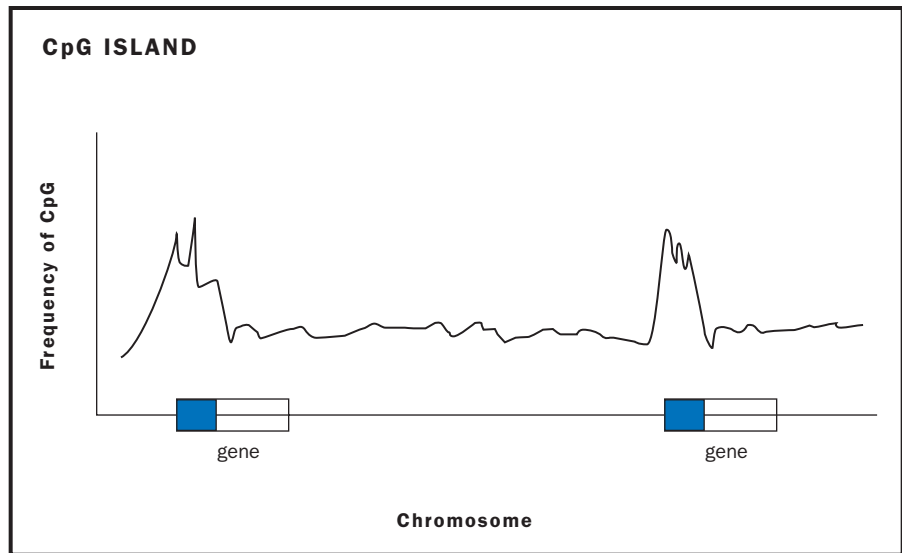
Methylation is the addition of a $-\text{CH}_3$ group at position 5 on the cytosine ring (box). Removal of the $-\text{NH}_2$ group by deamination converts this nucleotide into thymine. Methylation occurs on cytosines linked to guanines at their 3′ end.

promoter DNA sequence to which RNA polymerase binds to begin transcription

endonucleases enzymes that cut DNA or RNA within the chain

transposon genetic element that moves within the genome

High densities of methylated cytosines occur in gene promoter regions.



methylation can limit the spread of infectious virus from cell to cell, and prevent the damaging spread of transposon sequences.

Gene Repression

The addition of methyl groups to DNA can repress, or silence, gene expression by leading to a more compact DNA structure that excludes the binding of most proteins. Because of this, regions of DNA that are heavily methylated are not usually accessible to the binding of proteins needed for gene expression, such as **transcription factors**. Transcription repression is also aided by proteins that specifically bind to methylated DNA and contribute to the more compact DNA structure. These methyl-binding proteins contain a methyl-binding domain (MBD) that specifically recognizes methylated DNA. MeCP2, which causes a genetic disorder known as Rett syndrome, is one of these methyl-binding proteins that can bind to a single methylated cytosine in DNA and prevent the binding of other proteins like transcription factors. If it appears in a gene promoter region, it can prevent transcription from occurring.

Gene Imprinting

Methylation can also function in the process of genomic imprinting, which is found in sexually reproducing species. During sexual reproduction, each parent contributes one allele for each gene to the offspring. Genomic imprinting is a difference in gene expression that depends on whether the gene allele originated from the mother or the father. This differential pattern of gene expression occurs as the result of differential methylation in the gene promoter. One example of an imprinted gene is the insulin-like growth factor II (*IGF2*) gene. There are two copies (or alleles) of this gene, but only one is expressed. For *IGF2*, the maternal allele is methylated and silent, whereas the paternal allele is unmethylated and expressed. The opposite situation may occur in other genes. In this way, only one copy of an imprinted gene is expressed, and this provides a mechanism for a cell to determine the parental origin of certain genes.

transcription factors
proteins that increase the rate of gene transcription

DNA Methylation and Human Disease

Changes in DNA methylation patterns have been implicated in the development and progression of many types of cancers. Additionally, defects in DNA methylation have been associated with several genetic diseases, including ICF (Immunodeficiency, Centromere Instability, and Facial Anomalies), Rett, and Fragile X syndromes, all of which result in variable degrees of mental retardation. This common effect on neurological function may result from the fact that DNA methylation occurs at high levels in the brain, and implies that the brain requires DNA methylation for proper development.

ICF syndrome is a rare recessive disease characterized by variable immunodeficiency, developmental delays, distinctive facial features, and mental retardation. In 1999 it was found that patients with ICF syndrome have mutations in the DNA methyltransferase gene *DNMT3B*, located on human chromosome 20q11. These mutations impair the function of *DNMT3B*, resulting in an overall reduction in DNA methylation, or hypomethylation. This, in turn, leads to destabilization of the **centromeres** of chromosomes 1, 9, and 16. The alteration in chromosome structure leaves these chromosomes susceptible to DNA breakage and possibly alters the expression of genes located in these regions.

Rett syndrome and Fragile X syndrome are other genetic disorders that result from a disruption in the function of methylated DNA. Rett patients, who are almost all young females, at first develop normally. Later on, however, they develop mental retardation, autism, and movement disorders. These patients have a mutation in the gene for the methyl-binding protein MeCP2. This protein usually represses gene expression by binding tightly to methylated DNA and causing repression.

Fragile X syndrome is the most common form of inherited mental retardation. Fragile X results from an increase in the number of CGG repeats in the promoter region of the *FMRI* gene on the X chromosome. When the number of repeated sequences reaches the 200 to 600 copy range, the repeat itself becomes very heavily methylated, leading to silencing of the *FMRI* gene. The critical importance of DNA methylation in mammalian development is obvious, given the diseases that result when this process is improperly regulated. SEE ALSO GENE EXPRESSION, OVERVIEW OF CONTROL; IMPRINTING; NUCLEOTIDE.

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centromeres regions of the chromosome linking chromatids

Microbiologist

A microbiologist studies living organisms that are invisible to the naked eye. Microbiologists study bacteria, fungi, and other one-celled organisms (microbes), as well as viruses. A microbiologist may study a single molecule isolated from a bacterium, or a complex ecosystem with many microbial species.

In the course of their work, microbiologists do a variety of tasks. These include inoculating microbes using sterile techniques, viewing microbes under the microscope, purifying DNA or other molecules from microbes, counting microbes, preparing sterile **media**, and developing experiments to better understand these organisms, often using the techniques of genetic analysis. Many microbiologists supervise other employees, and they frequently work with a group of people to plan experiments or validate laboratory procedures. Some microbiologists teach or are engaged in sales, so their duties may include presenting information. As they gain experience, microbiologists working in a laboratory frequently assume more responsibility to supervise others and to plan a research program.

In most cases, microbiologists have training beyond high school. This training could consist of a two-year associate degree with coursework in biology, chemistry, physics, mathematics, and microbiology. Many microbiologists have a four-year bachelor's degree in microbiology, biology, or chemistry. This degree includes coursework in all the sciences, as well as several courses in microbiology. A master's degree usually requires an additional two years of coursework and research, and prepares individuals for additional supervisory or teaching positions. A doctoral degree (Ph.D.) typically requires an additional four to seven years of schooling, and opens up a wide range of teaching, research, and executive opportunities. In many cases, microbiologists who earn a Ph.D. do a further two to four years of postdoctoral research to gain additional research experience before they assume an independent position.

Some microbiologists work in hospital or clinical laboratories, where they identify disease-causing bacteria. Others work in the food industry, checking to make sure that food products do not contain **pathogenic** organisms. Environmental microbiologists study the role of bacteria and other organisms in ecological systems. Some microbiologists work in research laboratories, helping to make fundamental discoveries about microbes. Microbiologists also teach about these organisms in community colleges and universities.

The rewards of a career in microbiology are many, and may vary depending upon the career path taken. Medical microbiologists gain satisfaction from the knowledge that they assist in helping to prevent and treat disease. Food microbiologists are critical for maintaining the safety of our food supply. Research microbiologists are rewarded by the knowledge that they may, in the course of their work, learn something that no one else has ever understood. Teaching microbiologists gain satisfaction in helping others learn about a discipline they find fascinating.

Starting salaries for microbiologists also vary a great deal, depending on whether a person is employed in industry or in the public sector. In general, positions in industry command higher salaries than positions in

media (bacteria) nutrient source

pathogenic disease-causing

educational institutions, and individuals with many years of experience can command salaries at the high end of the range offered in their particular career path and at their educational level. Salaries for microbiologists with a bachelor's degree start at about \$18,000 per year, and range up to \$50,000 per year. For microbiologists with a master's degree, the starting salary is typically \$25,000 per year, and can rise to \$80,000. For microbiologists with a doctoral degree, the starting salary is about \$35,000 per year, and can range up to \$200,000 or more. SEE ALSO BIOTECHNOLOGY ENTREPRENEUR; COLLEGE PROFESSOR; LABORATORY TECHNICIAN; MOLECULAR BIOLOGIST.

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Mitochondrial Diseases

Mitochondria are intracellular organelles that play a critical role in cellular metabolism. Mitochondria contain the electron transport chain, which transfers electrons to oxygen by means of a process called oxidative **phosphorylation**. This process releases energy for the production of adenosine triphosphate (ATP) by forming a pH and electrical **gradient** (called the chemiosmotic gradient) across the inner mitochondrial membrane. In addition to oxidative phosphorylation, the mitochondria fulfill a number of other functions, including the following:

- Make ATP for cellular energy
- Metabolize fats, carbohydrates, and amino acids
- Interconvert carbohydrates, fats, and amino acids
- Synthesize some proteins
- Reproduce themselves (replicate)
- Participate in apoptosis
- Make free radicals

Of these functions **apoptosis** is particularly important in development and disease. However, human disease may result from impairment of any of these functions.

Mitochondria are inherited from the mother, but not from the father. In the process of egg formation, there is thought to be a “bottleneck” in mitochondrial number, such that the unfertilized egg may have as few as 1,000 mitochondria. This number increases 100-fold after the ovum is fertilized. The mitochondria contain their own DNA, mitochondrial or mtDNA, and during development there may be selective **amplification** of some of these mtDNA molecules, leading to increases or decreases in the presence of mutated mtDNAs.

phosphorylation addition of the phosphate group PO_4^{3-}

gradient a difference in concentration between two regions

apoptosis programmed cell death

amplification multiplication



ELECTRON TRANSPORT CHAIN DEFECTS OF LEIGH SYNDROME

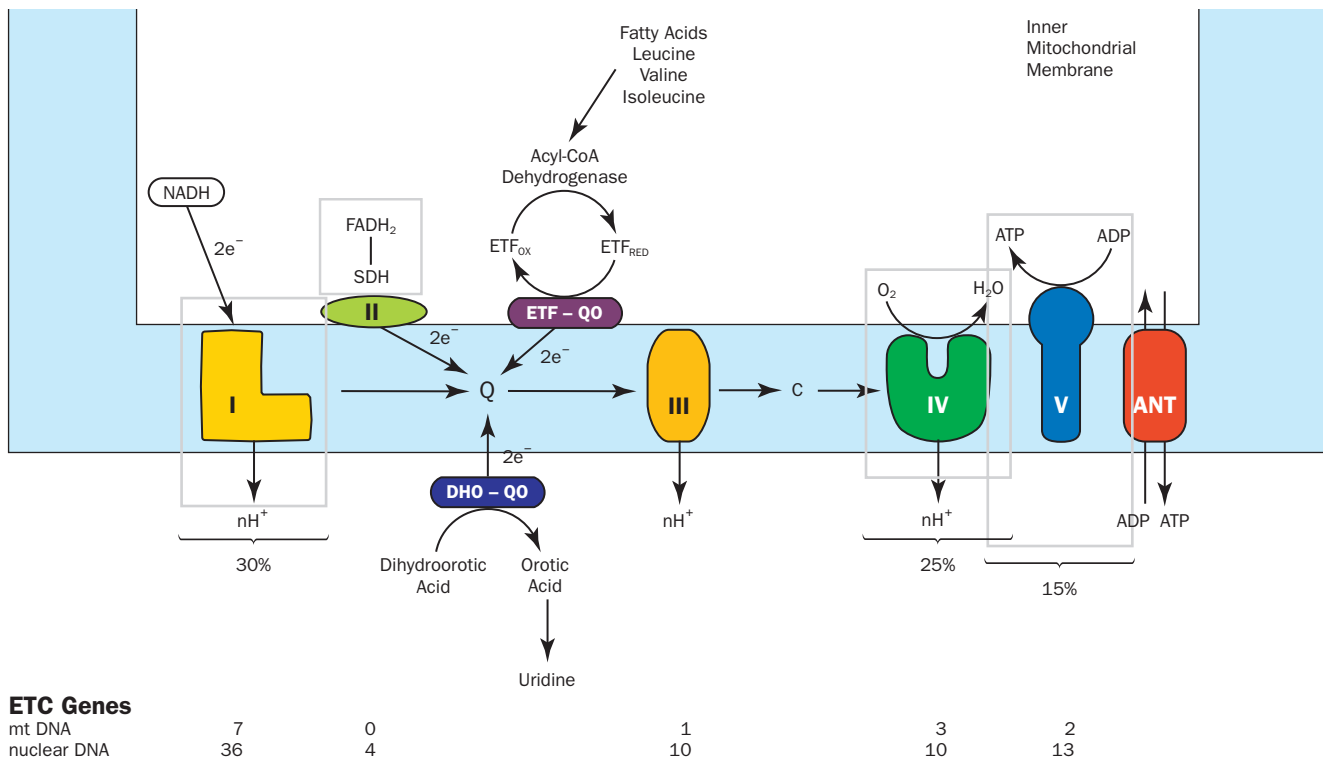


Figure 1. This figure provides an overview of the electron transport chain (ETC). Three defects causing Leigh's disease are highlighted above (purple boxes), with the percentage of cases due to these defects indicated. Numbers of nuclear and mitochondrial DNA (mtDNA) genes encoding the human ETC complexes are listed.

symbiotic describes a close relationship between two species in which at least one benefits

anaerobic without oxygen or not requiring oxygen

glycolysis the breakdown of the six-carbon carbohydrates glucose and fructose

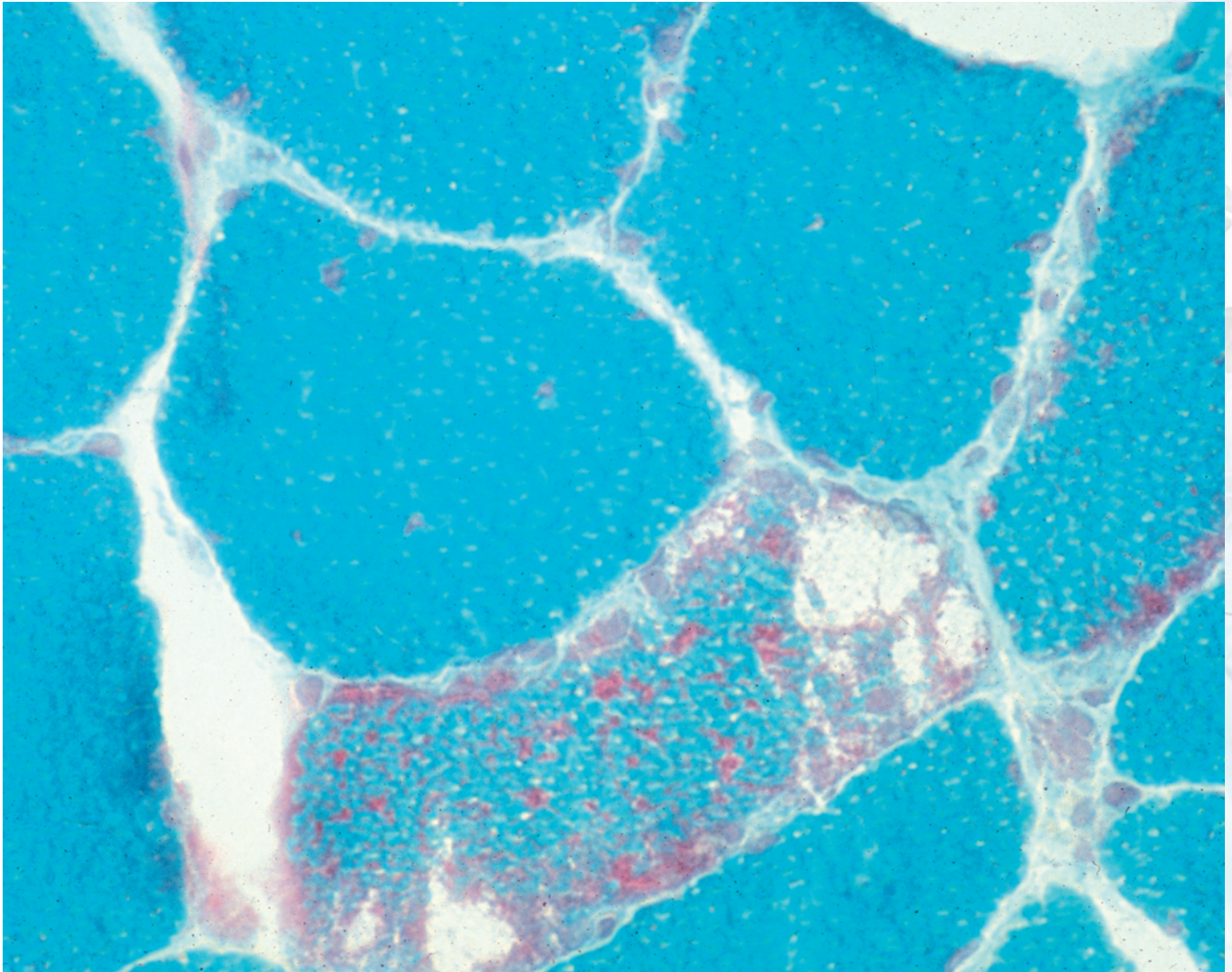
The Importance of the Electron Transport Chain

The origins of mitochondria are unknown, but the likely explanation, called the endosymbiont hypothesis, holds that they arose as free-living bacteria that colonized proto-eukaryotic cells, thereby establishing a **symbiotic** relationship. Primitive eukaryotic cells with intracellular mitochondria capable of metabolizing oxygen would have had an advantage in an oxygen-rich environment. The electron transport chain produces far more energy for each molecule of glucose consumed than is produced by **anaerobic** respiration. The oxidative phosphorylation process conducted by the mitochondria produces thirty-eight molecules of ATP, compared to two molecules of ATP produced by anaerobic **glycolysis**. Oxidative phosphorylation allows the conversion of toxic oxygen to water, a protective biological advantage.

A disadvantage of oxidative phosphorylation, however, is the formation of reactive oxygen species, such as singlet oxygen and hydroxyl radicals, which damage such cellular components as lipids, proteins, and DNA. A normally functioning electron transport chain produces reactive oxygen species from about 2 percent of the electrons that it transports. In disease states and in aging, larger quantities of reactive oxygen species are generated, and this may be a significant factor in cellular deterioration as well as a major contributor to the aging process.

Mitochondrial Genes and Disease

Mitochondrial DNA encodes approximately 3 percent of mitochondrial proteins. The relative contribution of the mitochondrial and nuclear genomes



in coding for electron transport chain subunits is detailed in Figure 1. Human mtDNA contains 16,569 nucleotide bases and encodes thirteen polypeptides of the electron transport chain, twenty-two transfer RNAs (tRNAs) and two ribosomal RNAs (rRNAs). In addition, mtDNA has a control region (termed the D-loop), which contains considerable genetic variation. The D-loop forms the basis of forensic medicine DNA identification and has been very useful in the molecular anthropological study of human origins.

In 1988 the first human disease associated with mtDNA deletions was reported. These patients suffered from muscle and brain diseases with ragged red fiber muscle disease (myopathy), with or without progressive neurological deterioration. Ragged red fibers are muscle fibers, that have a disorganized structure and an excess of abnormal mitochondria and that stain red when treated with a histochemical stain called modified Gomori trichrome (Figure 2). In 1988 Kearns-Sayre syndrome, which primarily affects the muscles, heart, and brain, was found to be due to mtDNA deletions or duplications. About the same time, the maternally inherited disorder Leber's hereditary optic atrophy was traced to point mutations in mitochondrial DNA encoding subunits of complex I of the electron transport chain.

Figure 2. Muscle biopsy of mitochondrial myopathy patient (modified Gomori trichrome stain) with ragged red fiber showing disorganization and mitochondrial proliferation. Mitochondria stain red. Other muscle fibers also have increased numbers of mitochondria.